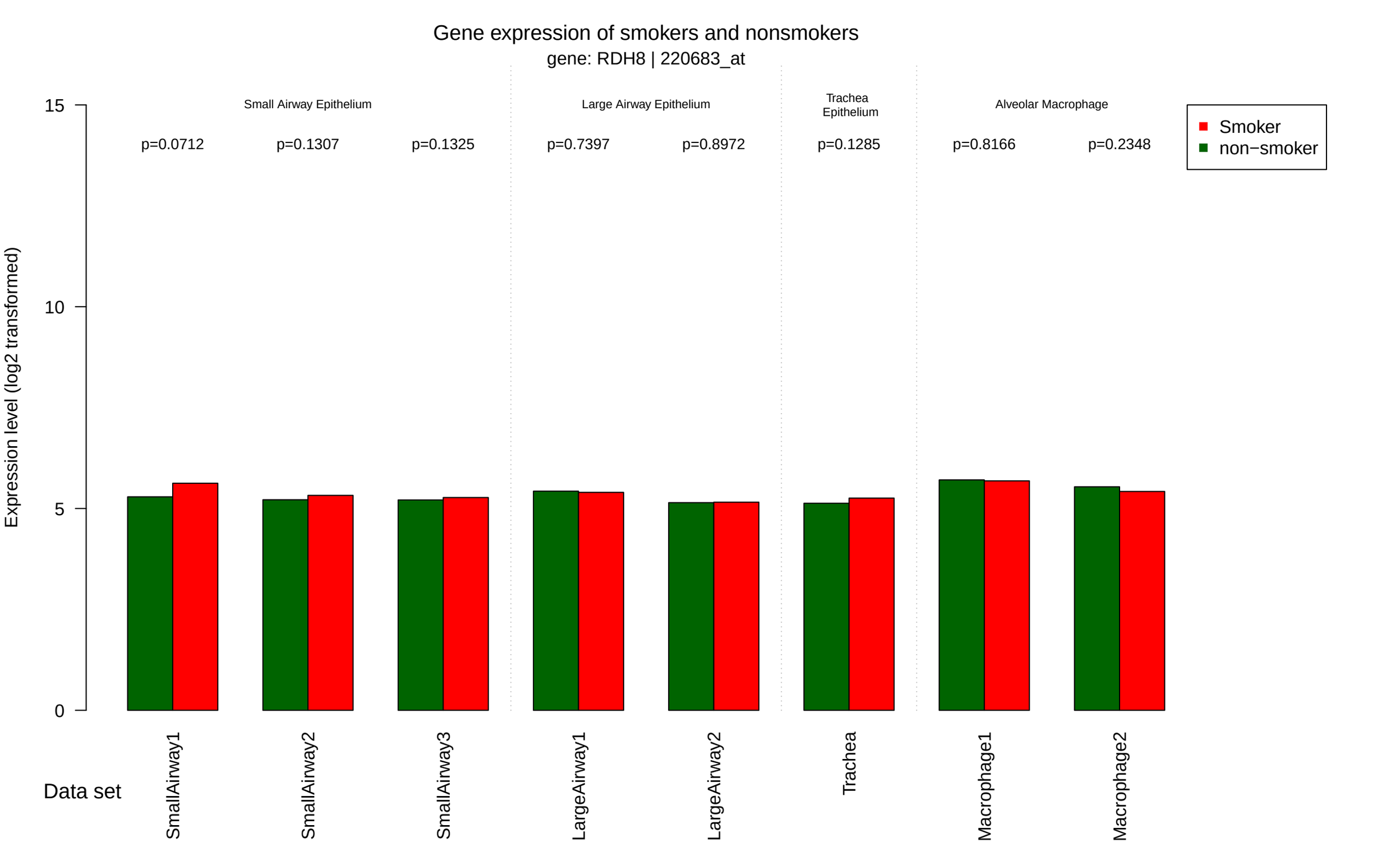
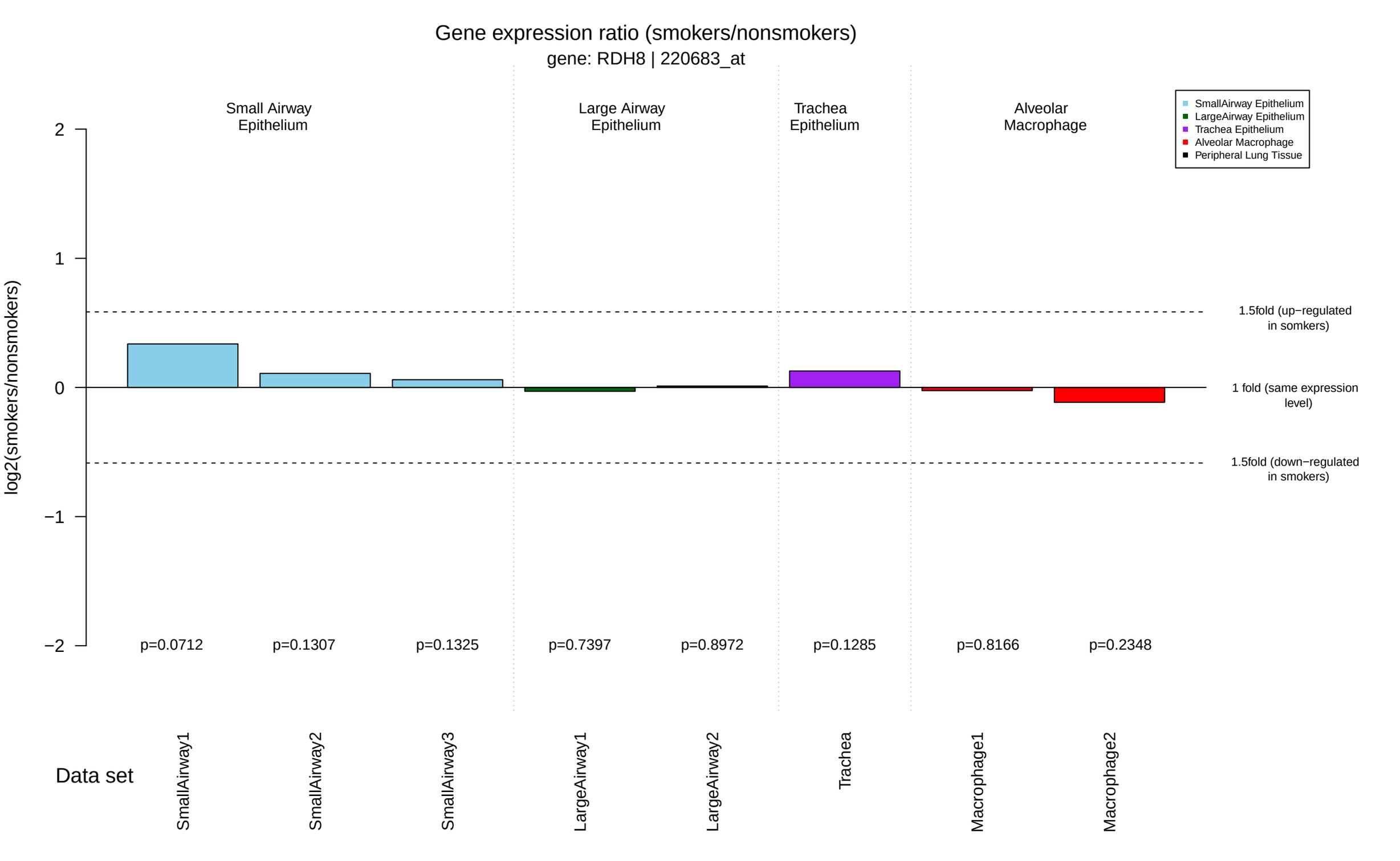
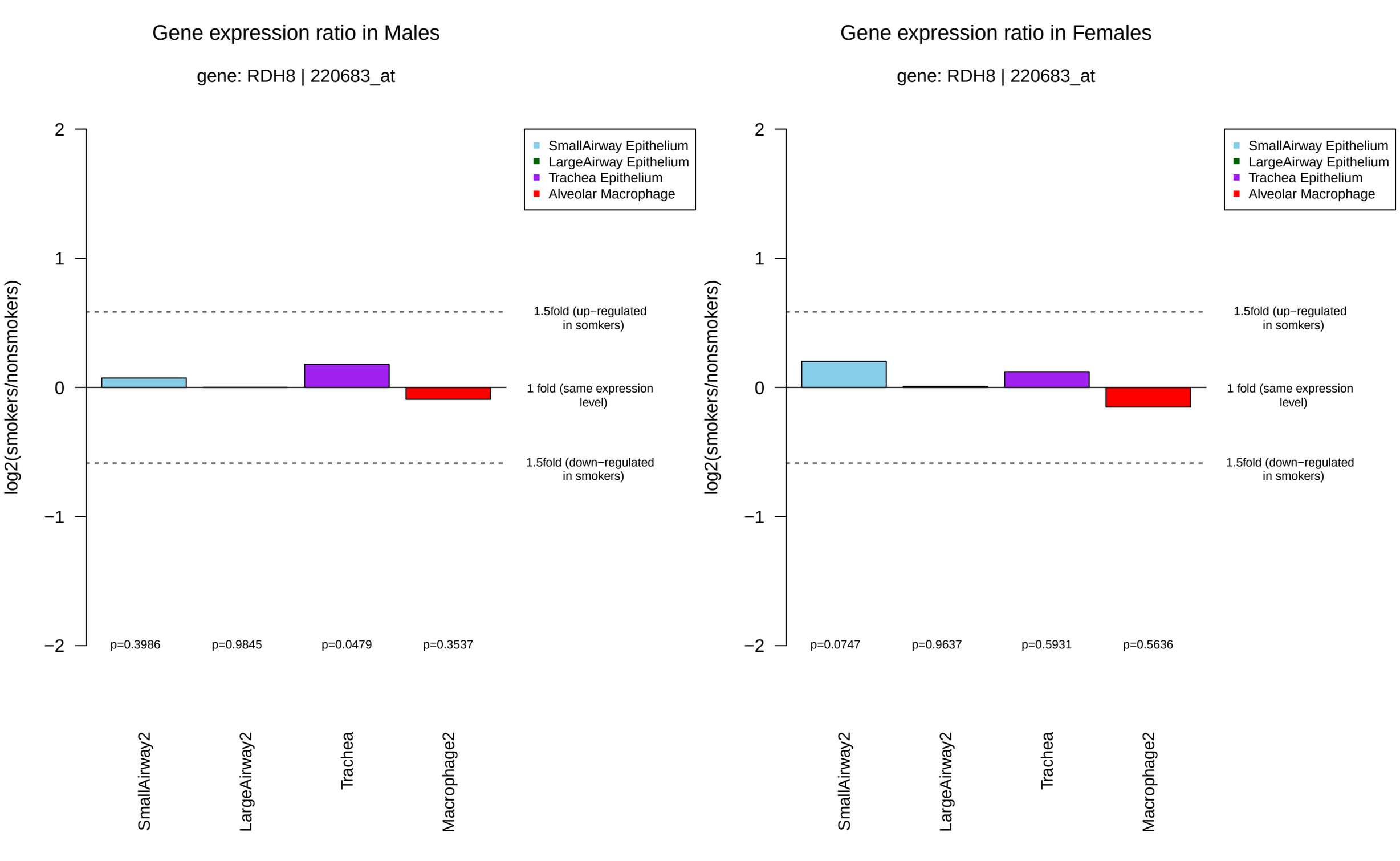
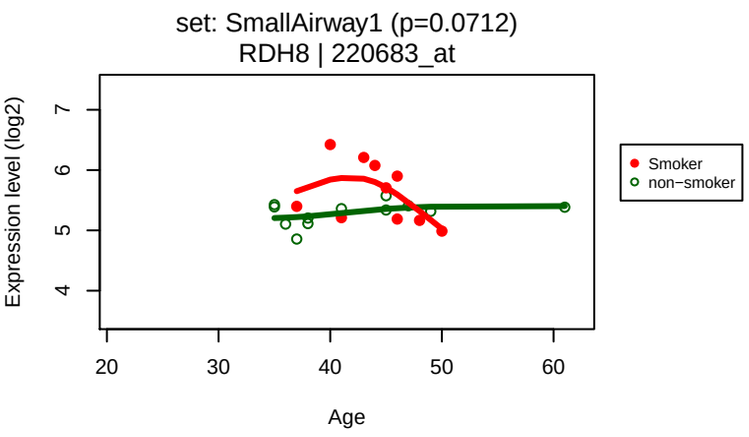
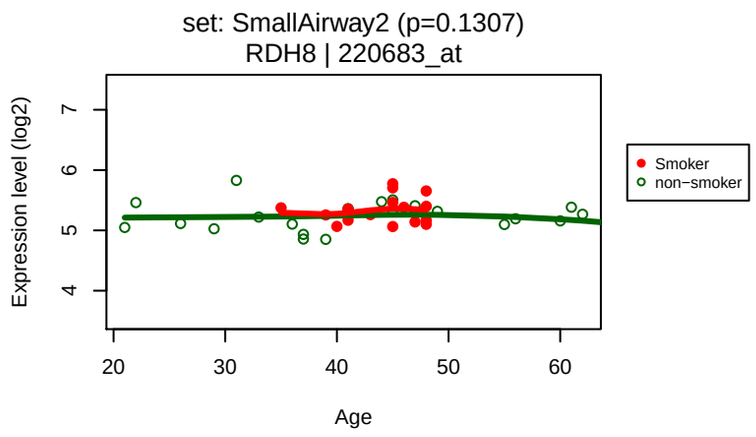
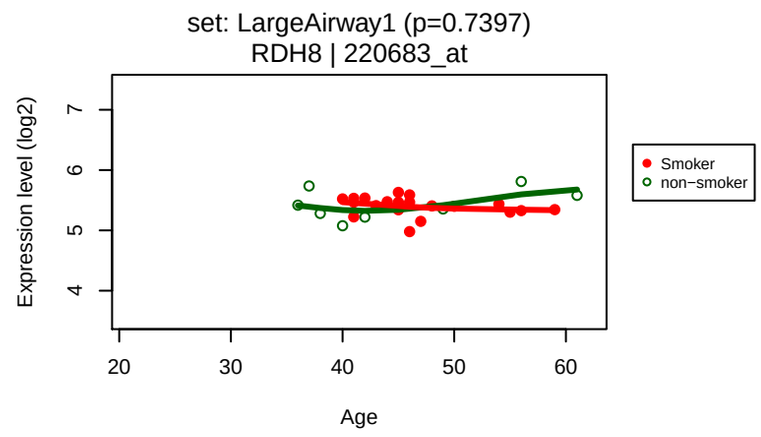

Gene expression of smokers and nonsmokers
gene: RDH8 | 220683_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0712
p=0.1307
p=0.1325
p=0.7397
p=0.8972
p=0.1285
p=0.8166
p=0.2348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

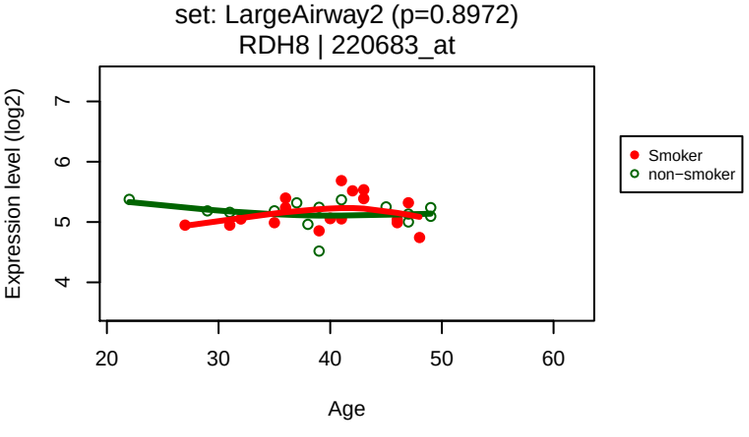

Gene expression ratio (smokers/nonsmokers)
gene: RDH8 | 220683_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0712
p=0.1307
p=0.1325
p=0.7397
p=0.8972
p=0.1285
p=0.8166
p=0.2348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

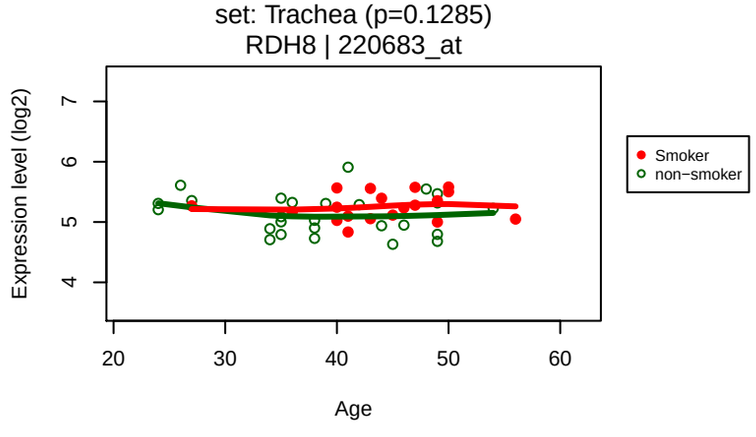

Gene expression ratio in Males
gene: RDH8 | 220683_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3986
p=0.9845
p=0.0479
p=0.3537
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH8 | 220683_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0747
p=0.9637
p=0.5931
p=0.5636
SmallAirway2
LargeAirway2
Trachea
Macrophage2

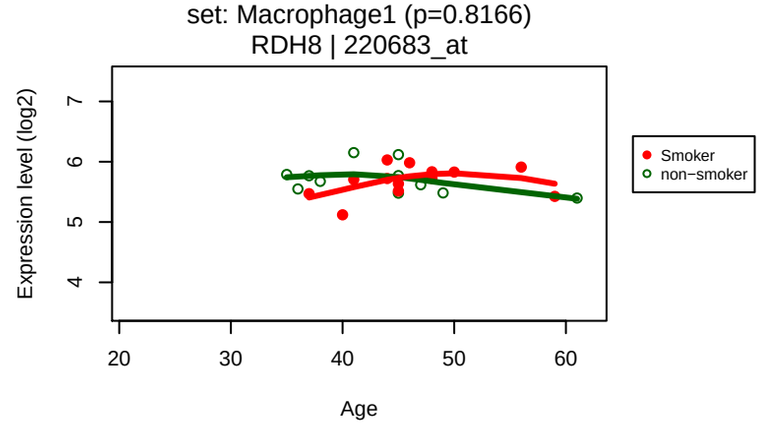

set: SmallAirway1 (p=0.0712)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker

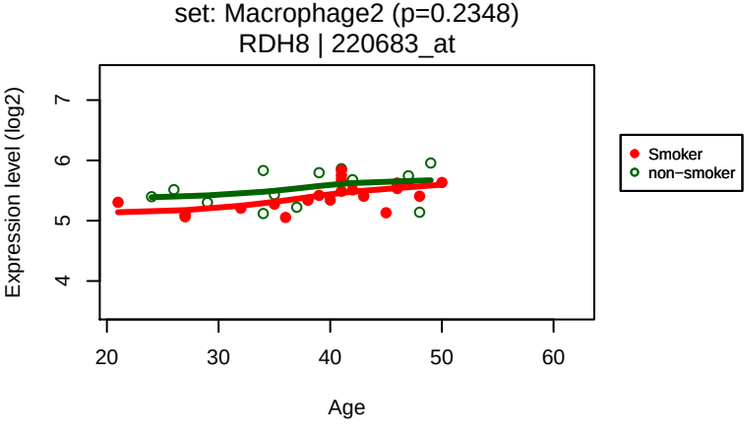

set: SmallAirway2 (p=0.1307)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker

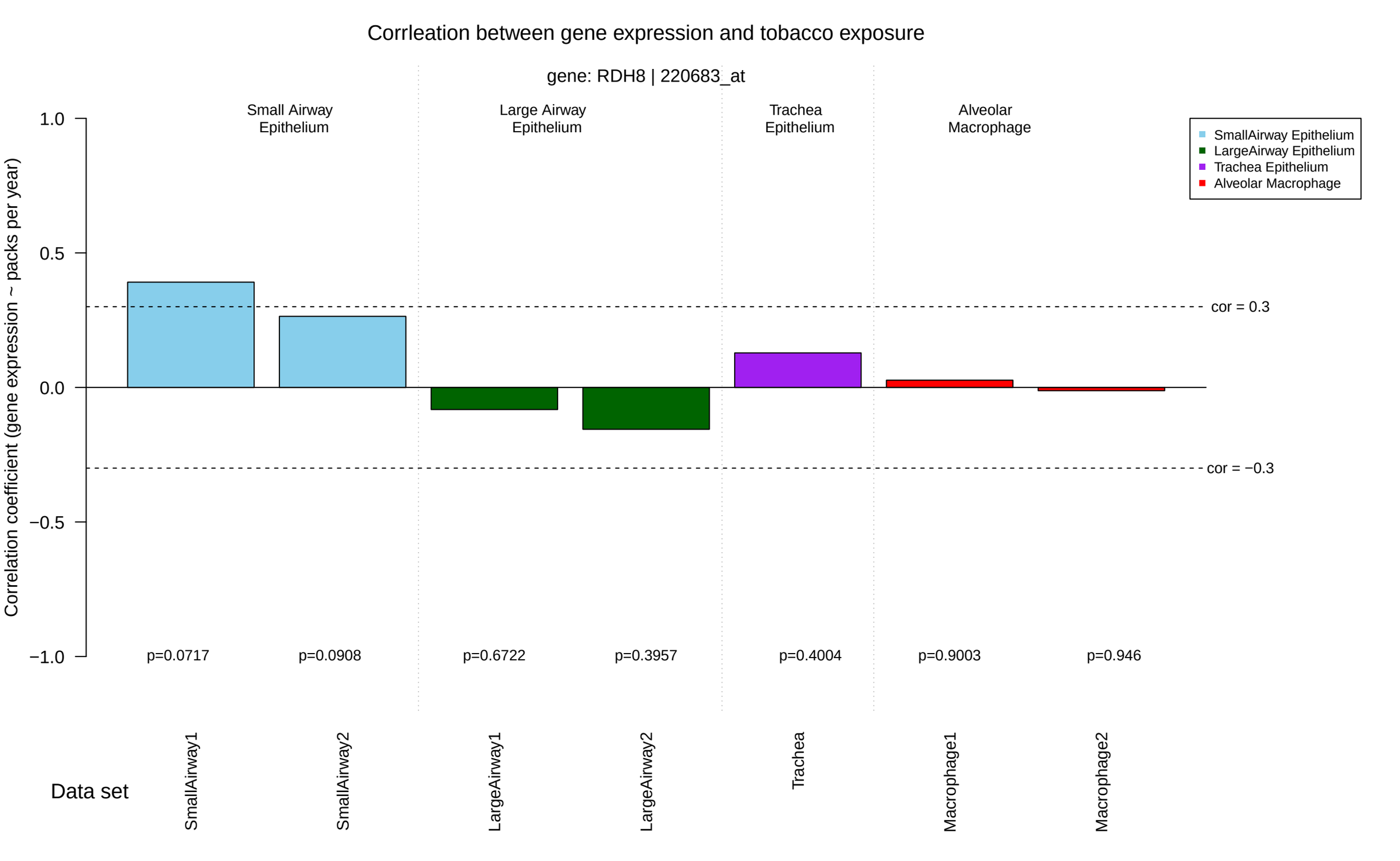

set: LargeAirway1 (p=0.7397)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8972)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1285)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8166)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2348)
RDH8 | 220683_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH8 | 220683_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0717
p=0.0908
p=0.6722
p=0.3957
p=0.4004
p=0.9003
p=0.946
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2